Midterm - Contribution Journal

John Grubbs

I loaded the healthcare_dataset.csv file, reviewed the rows, columns, data types, descriptive statistics, missing values, categorical values, and Test Results distribution.

I compared the billing distributions and skewness before and after the transformations. I encoded the target using LabelEncoder, removed non-predictive columns, and used one-hot encoding for categorical features. I created visualizations for the target distribution, medical condition and gender results, age, billing amounts, blood types, scaling, and transformation comparisons.

I checked the final processed data for missing values and displayed sample rows to verify the results. I also went over the work of my teammates to provide a second set of eyes.

Keira Burdine

My contribution to this project was removing duplicate records from the training dataset using Python in Jupyter Notebook. After splitting the original healthcare dataset into training and testing sets, I focused on cleaning the training data to improve its quality and reliability. I used the `drop_duplicates()` function in pandas to identify and remove duplicate rows without manually changing the original CSV file. Before removing duplicates, I recorded the number of rows in the training dataset, then compared it with the number of rows remaining after cleanup. This process helped me understand how duplicate records can affect data analysis and potentially influence the performance of a machine learning model. By performing this step through Python code, I followed the project requirement that all dataset modifications must be completed programmatically.

Through this contribution, I learned the importance of data cleaning as an essential step in preparing data for machine learning. Removing duplicate records helps reduce unnecessary repetition and allows the model to learn from unique observations. I also learned the importance of checking the dataset after preprocessing to confirm that the duplicates were successfully removed. In addition, I ensured that the testing dataset remained untouched during this cleaning step, helping preserve a fair evaluation of future machine learning models. Overall, this task strengthened my skills in Python, pandas, and data preprocessing while helping me understand how proper dataset preparation contributes to building more reliable machine learning models.

Luis Adrian

My contribution focused on feature engineering and the scal numericals feature. I created the Age_Group variable from Age using four predefined categories: under 18, 18–39, 40–59, and 60+. The original Age variable was retained to preserve its numerical information. I also displayed sample rows to check that the ages were assigned to the correct groups.

I also worked on preparing and exporting the final dataset. This included combining the processed training features with the encoded Test Results target, checking the final dimensions, and confirming that no missing values remained. The processed training dataset was saved as "`healthcare_train_processed.csv`", while the testing dataset was saved separately without preprocessing.